

DESIGN TASK SIMULATION :

QUESTION : Propose a data flow model for cloud-based radio resource allocation.

CODE IMPLEMENTATION :

```
clc;
clear;
close all;
% System parameters
numUsers = 10;
totalRB = 100;
% Generate random user demand and priority
demand = randi([5 20], 1, numUsers);
priority = randi([1 3], 1, numUsers);
% Initialize allocation
allocation = zeros(1, numUsers);
remainingRB = totalRB;
% Sort users according to priority
[~, index] = sort(priority, 'descend');
% Resource allocation
for i = 1:numUsers
    u = index(i);
    if remainingRB >= demand(u)
        allocation(u) = demand(u);
    else
        allocation(u) = remainingRB;
    end
    remainingRB = remainingRB - allocation(u);
    if remainingRB == 0
        break;
    end
end
end
```

```

% Display results
disp('User Demand Priority Allocation');
disp([(1:numUsers)' demand' priority' allocation']);

% Plot results
figure;
subplot(1,2,1);
bar([demand' allocation'], 'grouped');
xlabel('Users');
ylabel('Resource Blocks');
title('Demand vs Allocation');
legend('Demand', 'Allocated');
grid on;
subplot(1,2,2);
pie([sum(allocation) remainingRB], {'Allocated','Unused'});
title('Resource Utilization');

```

OUTPUT :

